

Thm: let $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$ be a f.d. rep'n of a linear
 Lie alg. for which $\rho(\mathfrak{g}) \subseteq \mathcal{L}(\mathfrak{g})$ consists of nilpotent
 operators. Then there is a basis β of V for which

$$[\rho(\mathfrak{g})]_{\beta} \subseteq \mathcal{L}_n^0(\mathbb{K})$$

Here $\rho(\mathfrak{g})$ is a nilpotent Lie algebra.

Pf: (2) We wish to find $v_0 \in V \setminus \{0\}$ s.t. $\rho(X)v_0 = 0, \forall X \in \mathfrak{g}$.

We shall proceed by induction on dimension.

First if $\mathfrak{g} = \mathbb{K} X_0$ (1-dim'l, hence abelian)

then $\rho(X_0)$ is nilpotent and admits $v_0 \in V \setminus \{0\}$ with $\rho(X_0)v_0 = 0$.

Now sps $d \geq 2$, and for any Lie algebra $\mathfrak{h}$ with $\dim_{\mathbb{K}} \mathfrak{h} < d$,

and for any Lie rep'n $\alpha: \mathfrak{h} \rightarrow \mathfrak{gl}(W)$ (W f.d.)

consisting of nilpotent operators, we have $w_0 \in W \setminus \{0\}$ so

$$\alpha(X)w_0 = 0 \text{ for } X \in \mathfrak{h}.$$

let $\tilde{\mathfrak{g}} = P(\mathfrak{g}) \subseteq \mathfrak{gl}(V)$, and $\mathfrak{h} \subseteq \tilde{\mathfrak{g}}$ be a proper Lie subalgebra

of maximal dimension, so $\dim_{\mathbb{K}} \mathfrak{h} < \dim_{\mathbb{K}} \tilde{\mathfrak{g}} \leq \dim_{\mathbb{K}} \mathfrak{g}$. Then we

have that $\text{ad}: \tilde{\mathfrak{g}} \rightarrow \mathfrak{gl}(\tilde{\mathfrak{g}})$ has that $\text{ad}/\mathfrak{h}$ admits $\mathfrak{h}$.

is an invariant subspace, and we consider

$$\alpha: \mathfrak{h} \rightarrow \mathfrak{gl}(\tilde{\mathfrak{g}}/\mathfrak{h}), \quad \alpha(X)(Y+\mathfrak{h}) = [X, Y] + \mathfrak{h}, \quad X \in \mathfrak{h}, Y \in \tilde{\mathfrak{g}}$$

Our assumption on $\tilde{\mathfrak{g}} = P(\mathfrak{g})$ (nilpotent operation), and Remark (ii)

above, we see that $\alpha(\mathfrak{h})$ consists of nilpotent operators.

Now, the inductive hypothesis provides $\lambda_0 + h \neq 0 + h$

i.e., $\lambda_0 \in \tilde{\mathfrak{g}} \setminus h$. so $\alpha(X)(\lambda_0 + h) = 0 + h$ for $X \in h$,

i.e., $[X, \lambda_0] \in h$, so by assumption of maximal dimension as a

subalgebra, we see that $\tilde{\mathfrak{g}} = h + k\lambda_0$. Moreover, $h \triangleleft \tilde{\mathfrak{g}}$
(the ideal)

Now let $W = \bigcap_{X \in h} \ker(X) \subseteq U$. If $w \in W$, $X \in h$.

$$\underbrace{\lambda_0}_{=0} w = \underbrace{\lambda_0 X w}_{=0} - \underbrace{[X, \lambda_0] w}_{\in h, =0} = 0 \Rightarrow X(W) \subseteq W.$$

As λ_0 is nilpotent (assumption on $\tilde{\mathfrak{g}}$), $\lambda_0|_W$ is nilpotent

in $L(W)$. By Remark (ii), this has 0 as eigenvalue.

i.e., there is $w \in (W \setminus \{0\}) \cap \ker \lambda_0$, and we see that

$$\lambda w = 0 \text{ for } \lambda \in \tilde{\mathfrak{g}}$$

(II). Now we find the desired basis.

Let $v_1 \in V \setminus \{0\}$, so $P(g)v_1 = \{0\}$, let

$$P_1 = \overline{P} : g \rightarrow \mathfrak{gl}(V/U_1), \text{ where } U_1 = \mathbb{K}v_1$$

Let $v_2 \in V \setminus U_1$ be so $P_1(g)(v_2 + U_1) = \{0 + U_1\}$

So V_2/U_1 is P_1 -invariant

Let $P_2 = \overline{P_1} : g \rightarrow \mathfrak{gl}(V/U_1)/(U_2/U_1) = \mathfrak{gl}(V/U_2)$

when $U_2 = \text{span}_{\mathbb{K}} \{v_1, v_2\}$

$\vdots$

Let $v_k \in V \setminus U_{k-1}$ be so $P_{k-1}(g)(v_k + U_{k-1}) = \{0 + U_{k-1}\}$.

Let $P_k = \overline{P_{k-1}} : g \rightarrow \mathfrak{gl}(V/U_{k-1})$, let $U_k = U_{k-1} + \mathbb{K}v_k$

This process stops at $k=n=\dim_k V$. The spaces

$V_1 \subseteq V_2 \subsetneq \dots \subsetneq V_n = V$ $\dim V_k/V_{k-1} = 1$ are constructed.

$P(g)V_1 = \{0\}$, $P(g)V_k \subseteq V_{k-1}$ for $k=2, \dots, n$. Let $\beta = (v_1, \dots, v_n)$,

and we have

$$[P(g)]_\beta \in \mathcal{L}_n(V) \quad [\text{check}]$$

Then for $k \in \mathbb{N}$,

$$C^k(P(g)) \subseteq C^k(\mathcal{L}_n(V)), \quad C^n(\mathcal{L}_n(V)) = \{0\}.$$

Corollary (Engel's Thm) let $\mathfrak{g}$ be a linear Lie algebra

Then

$\mathfrak{g}$ is nilpotent $\Leftrightarrow \text{ad}(g) \in \mathcal{L}(\mathfrak{g})$ consists of
nilpotent operators

Lie's Thm: let $\mathfrak{g} \in \mathfrak{gl}_n(\mathbb{C})$ be a solvable $\mathbb{K}$ -Lie algebra

($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$).

(i) Then there are v_0 in $\mathbb{C}^n \setminus \{0\}$ and a $\mathbb{K}$ -linear $\lambda: \mathfrak{g} \rightarrow \mathbb{C}$ for which $Xv_0 = \lambda(X)v_0$ for X in $\mathfrak{g}$ ("eigenfunction")

(ii) Then there is g in $GL_n(\mathbb{C})$ for which

$$g \mathfrak{g} g^{-1} \subseteq \mathfrak{t}_n(\mathbb{C}) \quad (\text{change of basis into upper triangles})$$

Pf: (i) (I) If $\mathfrak{h} \subseteq \mathfrak{g}$ ($\mathbb{K}$ -Lie subalg), then $\mathfrak{h}$ is solvable

as $\mathcal{D}^k(\mathfrak{h}) \subseteq \mathcal{D}^k(\mathfrak{g})$. Also, if $\mathfrak{h} \neq \{0\}$, then

$\dim_{\mathbb{K}} \mathcal{D}(\mathfrak{h}) < \dim_{\mathbb{K}} \mathcal{D}(\mathfrak{g})$. So there is a subspace

$\mathfrak{k} \subseteq \mathfrak{h}$ with

$$\mathcal{D}(\mathfrak{h}) \subseteq \mathfrak{k} \subseteq \mathfrak{h} \quad \text{and} \quad \dim_{\mathbb{K}} \mathfrak{k} + 1 = \dim_{\mathbb{K}} \mathfrak{h}.$$

But the $[h, k] \subseteq [h, h] = \mathcal{D}(h) \subseteq k$.

So $k \triangleleft h$ (Lie ideal), in particular, k is
a Lie subalgebra of $\mathfrak{g}$.

Hence, we see that $\mathfrak{g}$ admits a Lie algebra of

each dimension $1, \dots, \dim_{\mathbb{K}} \mathfrak{g} - 1, \dim_{\mathbb{K}} \mathfrak{g}$
 $\quad \quad \quad \uparrow$
 $\quad \quad \quad \mathfrak{g}$ itself.

(II). (Induction on dimension of subalgebra)

If $h = \mathbb{K}X_0$, $X_0 \in \mathfrak{g} \setminus \{0\}$, $w_0 \in \mathbb{C}^n \setminus \{0\}$ is any

eigenvector of X_0 , with eigenvalue λ_0 , then $\lambda: h = \mathbb{K}X_0 \rightarrow \mathbb{C}$,

$\lambda(\alpha X_0) = \alpha \lambda_0$ does the job.

Hence we fix $1 \leq d \leq \dim_{\mathbb{K}} \mathfrak{g}$ and assume that for any
subalg. h of $\mathfrak{g}$ of $\dim_{\mathbb{K}} h = d$, and any proper subalg. k of h ,

there exists a $w_0 \in \mathbb{C}^n \setminus \{0\}$, and an eigenfunction

$$\lambda: k \rightarrow \mathbb{C} \text{ (}\mathbb{C}\text{-linear) so}$$

$$Y w_0 = \lambda(Y) w_0, \quad Y \in k.$$

We may, and will, assume that $\dim_{\mathbb{C}} k = d-1$, $\forall w \in k$.

Thus, fix $\lambda_0 \in h \setminus k$, and let k be the largest integer s.t.

$$w_0, w_1 := \lambda_0 w_0, \dots, w_k = \lambda_0^k w_0 \text{ are } \mathbb{C}\text{-lin. indep.}$$

We set $W_{-1} = \{0\}$, $W_j = \text{span}_{\mathbb{C}} \{w_0, \dots, w_j\}$, $j = 0, 1, \dots, k$.

and hence $\lambda_0(W_{j-1}) \subseteq W_j$ for $j = 0, \dots, k-1$, while $\lambda_0(W_k) \subseteq W_k$

(II a) let's see for $Y \in k$ that

$$Y(W_j) \subseteq W_j, \quad Y w_j + W_{j-1} = \lambda(Y) w_j + W_{j-1}, \quad j = 0, 1, \dots, k.$$

The case $j=0$ is given by choice of w_0 and W_{-1} .

If the result is true for $i=1, \dots, j-1$, we have

$$Yw_j + W_{j-1} = YX_0w_{j-1} + W_{j-1}$$

$$\begin{aligned} &= \lambda([X_0, Y])w_{j-1} + W_{j-2} \in W_{j-1} \\ &= \underbrace{X_0 Y w_{j-1}} - \underbrace{[X_0, Y] w_{j-1}} + W_{j-1} \\ &= \lambda(Y)w_{j-1} + W_{j-2} \end{aligned}$$

$$= \lambda(Y)X_0w_{j-1} + W_{j-1}$$

$$= \lambda(Y)w_j + W_{j-1}$$

and then $Y(W_j) \subseteq W_j$.

Thus if $\beta = (w_0, \dots, w_k)$, which is a base for W_k ,

then we see for $Y \in \mathfrak{k}$.

$$[Y|_{W_k}]_{\beta} = \begin{bmatrix} \lambda(Y) & & * \\ & \ddots & \\ 0 & & \lambda(Y) \end{bmatrix}$$

(II b). We shall see for $Y \in \mathcal{K}$, that $Y|_{W_k} = \lambda(Y) I_{W_k}$.

(i.e., diagonal instead of upper triangular)

From (II a), above, we see that

$$E = E^2 \text{ in } M_n(\mathbb{C}), \quad E(\mathbb{C}^n) = W_k.$$

$$\text{Tr}(YE) = \text{Tr}(Y|_{W_k}) = (k+1)\lambda(Y). \text{ Since } [X_0, Y] \in \mathcal{K}$$

for $Y \in \mathcal{K}$ (by (I)), we have

$$\text{Tr}([X_0, Y]|_{W_k}) = \text{Tr}((X_0 Y - Y X_0)E) \quad (†)$$

Since Y, X_0 have W_k as an invariant subspace.

$$X_0 E = E X_0 E, \quad Y E = E Y E$$

$$\text{So } (4) = \text{Tr} (X_0 \bar{E} Y E - Y \bar{E} X_0 E)$$

$$= \text{Tr} ([X_0 \bar{E}, Y E])$$

$$= \text{Tr} ([X_0|_{m_k}, Y|_{m_k}]) = 0$$

We conclude that $\lambda(\bar{X}, Y) = 0$.

Now $Y u_0 = \lambda(Y) u_0$ (choice of u_0), and we assume

$$Y u_{j-1} = \lambda(Y) u_{j-1}$$

$$\begin{aligned} \text{Then } Y u_j &= Y X_0 u_{j-1} = X_0 Y u_{j-1} - [\bar{X}_0, Y] u_{j-1} \\ &= X_0 \lambda(Y) u_{j-1} - \underbrace{\lambda(\bar{X}_0, Y)}_{=0} u_{j-1} \end{aligned}$$

$$= \lambda(Y) X_0 u_{j-1}$$

$$= \lambda(Y) u_j$$

Hence $Y u_j = \lambda(Y) u_j$ for all $j=0, \dots, k$.

(II c). Now let $v_0 \in W_k \setminus \{0\}$ be any eigenvector for X_0 ,
eigenvalue λ_0 . Define

$$\mu: h = kX_0 + l \rightarrow \mathbb{C}, \mu(\alpha X_0 + Y) = \alpha \lambda_0 + \lambda(Y)$$

and we see that $Xv_0 = \mu(X)v_0$ for $X \in h$.

We do this for $h=g$ and get (i).

(ii). Let $v_1 \in \mathbb{C}^n \setminus \{0\}$, $\lambda_2: g \rightarrow \mathbb{C}$ be so $Xv_1 = \lambda_2(X)v_1$ for $X \in g$.

Let $V_1 = \mathbb{C}v_1$ so $g(V_1) \subseteq V_1$, define

$$P_1: g \rightarrow \mathfrak{gl}(\mathbb{C}^n/V_1) \text{ by } P_1(X)(v+V_1) = Xv+V_1.$$

Then $P_1(g)$ is solvable, $D^k(P_1(g)) \subseteq P_1(D^k(g))$, $\forall k \in \mathbb{N}$.

Let $v_2 \in \mathbb{C}^n \setminus V_1$, so $v_2+V_1 \neq 0+V_1$, and

$$\lambda_2: P_1(g) \rightarrow \mathbb{C} \text{ be so } P_1(X)(v_2+V_1) = \lambda_2(X)v_2+V_1.$$

let $V_2 = \text{span}_{\mathbb{C}} \{v_1, v_2\}$ so $g(V_2) \subseteq V_2$, and we

can define $P_2: g \rightarrow gl(\mathbb{C}^n/V_2)$, $P_2(X)(v+V_2) = Xv + V_2$

We continue to find linearly indep. $v_1, \dots, v_n$ so

$$V_k = \text{span}_{\mathbb{C}} \{v_1, \dots, v_k\}, g(V_k) \subseteq V_k,$$

$$P_k(X)(v_k + V_{k-1}) = \lambda_k(X) v_k + V_{k-1}.$$

Hence we let $J \in GL_n(\mathbb{C})$ be similarity exchanging

standard basis and $(v_1, \dots, v_n)$, so

$$J g J^{-1} \subseteq L_n(\mathbb{C}).$$

□

Corollary (to Lie's Thm):

let $\mathfrak{g}$ be a linear Lie algebra. Then

$$\mathfrak{g} \text{ is solvable} \Leftrightarrow [\mathfrak{g}, \mathfrak{g}] \text{ is nilpotent}$$

Pf: ($\Rightarrow$) We assume that $\mathfrak{g} \subseteq \mathfrak{gl}(V)$, V $\mathbb{C}$ -v.s., and there is

a basis β for so $[\mathfrak{g}]_{\beta} \subseteq \mathfrak{gl}_n(\mathbb{C})$, $n = \dim_{\mathbb{C}} V$ (Lie's Thm)

Thus, $[[\mathfrak{g}, \mathfrak{g}]]_{\beta} \subseteq \mathfrak{gl}_n^{\circ}(\mathbb{C})$, so $C^n(\mathfrak{g}) = \{0\}$.

($\Leftarrow$): $[\mathfrak{g}, \mathfrak{g}]$ nilpotent $\Rightarrow [\mathfrak{g}, \mathfrak{g}]$ solvable $\Rightarrow \mathfrak{g}$ solvable.

□.

Cartan's Thm: let $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{K})$ be a matrix Lie algebra.

if $\text{Tr}(XY) = 0$ for X, Y in $\mathfrak{g}$, then $\mathfrak{g}$ is solvable

Pf: We will show that $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.

By Engel's Thm, it suffices to show that

$\text{ad}([g, g]) \in L([g, g])$ consists of nilpotent operators.

if $K = \mathbb{R}$, we set $g \in \mathfrak{gl}_n(\mathbb{R}) \subseteq \mathfrak{gl}_n(\mathbb{C})$, and we work in $\mathfrak{gl}_n(\mathbb{C})$.

In this case, replace g by $g_{\mathbb{C}} = g + ig$, and we'll show $[g_{\mathbb{C}}, g_{\mathbb{C}}]$ is nilpotent. We hence assume g is $\mathbb{C}$ -linear algebra.

Hence we fix $X \in [g, g]$ so

$$\bullet X = \sum_{j=1}^m [X_j, Y_j], \quad X_j, Y_j \in g.$$

• (Jordan decomposition)

$$X = D + N = g^{-1} \left(\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} + N' \right) g, \quad g \in GL_n(\mathbb{C})$$

$[D, N] = 0$ since $D = p_D(X)$ (p_D a(n associated) $\mathbb{C}$ -polynomial)

We let $F_{ij} = g E_{ij} g^{-1}$ (matrix units), and in $L(\mathfrak{gl}_n(\mathbb{C}))$

- $\text{ad}(D) F_{ij} = (\lambda_i - \lambda_j) F_{ij}$

So $\text{ad}(D)$ is diagonalizable. Also, $\text{ad}(N)$ is nilpotent in $L(\mathfrak{gl}_n(\mathbb{C}))$

(Use Remark (ii) before Engel's Thm), and $[\text{ad}(D), \text{ad}(N)] = 0$.

Hence, by uniqueness of Jordan Decomposition there is a

$\mathbb{C}$ -polynomial P_D so $\text{ad}(D) = P_D(\text{ad}(X))$.

Let f, Q be $\mathbb{C}$ -polyc st.

$$f(\lambda_j) = \overline{\lambda_j}, \quad Q(\lambda_i - \lambda_j) = \overline{\lambda_i} - \overline{\lambda_j}, \quad i, j = 1, \dots, n.$$

Then $f(\lambda_i) - f(\lambda_j) = Q(\lambda_i - \lambda_j)$, and thus

$$\text{ad } f(D) = Q(\text{ad}(D)) = Q \circ P_D(\text{ad}(X)).$$

This implies $\text{ad } f(D)(\mathfrak{g}) \subseteq \mathfrak{g}$, as $\text{ad}(X)\mathfrak{g} \subseteq \mathfrak{g}$, and $\mathfrak{g}$ is a $\mathbb{C}$ v.s.

$$\begin{aligned}
\text{Then } \operatorname{Tr}(f(D)X) &= \sum_{j=1}^m \operatorname{Tr}(f(D)(X_j Y_j - Y_j X_j)) \\
&= \sum_{j=1}^m \operatorname{Tr}(f(D)X_j Y_j - X_j f(D)Y_j) \\
&= \sum_{j=1}^m \operatorname{Tr}(\underbrace{(ad f(D)X_j)}_{\in \mathfrak{g}} \underbrace{Y_j}_{\in \mathfrak{g}}) = 0, \text{ by assumption.}
\end{aligned}$$

Since $[D, N] = 0$, $[f(D), N] = 0$ so $f(D)N$ is nilpotent.

Thus

$$\begin{aligned}
\operatorname{Tr}(f(D)X) &= \operatorname{Tr}(f(D)D) + \underbrace{\operatorname{Tr}(f(D)N)}_{=0} \\
&= \sum_{i=1}^n |\lambda_i|^2 \cdot f f(D) D f^{-1} = [\lambda_1 \dots \lambda_n] [\lambda_1 \dots \lambda_n]
\end{aligned}$$

We conclude that each eigenvalue of X is 0, and thus (by Jordan form),

X is nilpotent. Thus $[g, g] \subseteq \mathfrak{gl}(\mathbb{C})$ is nilpotent, by Engel's Thm.

□

Quotient algebra: if $\mathfrak{g}$ is a linear Lie algebra, $\mathfrak{i} \triangleleft \mathfrak{g}$ (Lie ideal),

then $\mathfrak{g}/\mathfrak{i}$ is a Lie algebra by

$$[X+\mathfrak{i}, Y+\mathfrak{i}] = [X, Y] + \mathfrak{i}$$

[Notice: Jacobi identity on $\mathfrak{g} \Rightarrow$ Jacobi identity on $\mathfrak{g}/\mathfrak{i}$].

Ado's Thm: If $\mathfrak{g}$ is a finite Lie algebra ($\mathbb{K}$ -v.s. with bilinear form

$[\cdot, \cdot]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ satisfying Jacobi identity), then there is

a Lie algebra HM $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}_n(\mathbb{K})$, $\ker \rho = \{0\}$.

Prop'n: let $\mathfrak{g}$ be a linear Lie algebra

(i) if $\mathfrak{i} \triangleleft \mathfrak{g}$, s.t. both $\mathfrak{i}$ and $\mathfrak{g}/\mathfrak{i}$ are solvable, then

$\mathfrak{g}$ is solvable.

(ii) if $i, j \triangleleft g$ are both solvable, so too is $i+j$

Pf: (i) Say $D^m(g/i) = \{0+i\}$, $D^n(i) = \{0\}$.

Then $D^m(g) \subseteq i$, so $D^{m+n}(g) \subseteq D^n(i) = \{0\}$.

(ii) $(i+j)/i \cong j/(i \cap j)$. Hence j solvable.

$\Rightarrow (i+j)/i$ solvable

$\Rightarrow i+j$ is solvable.

$\square$.

Def'n: If g is a linear Lie algebra, then its solvable radical

$$\text{rad}(g) = \text{span}_{\mathbb{K}} \{i: i \triangleleft g \text{ is a solvable ideal}\}$$

If $\{x_1, \dots, x_m\}$ is a max'l lin. indep. subset of $\text{rad}(g)$,

we may find i_k , a solvable ideal of g , so $x_k \in i_k$.

Then $\text{rad}(\mathfrak{g}) = \mathfrak{z} + \dots + \mathfrak{z}_n$ and hence is solvable.

The Killing Form

Recall: $\text{Tr}: M_n(\mathbb{K}) \rightarrow \mathbb{K}$, $\text{Tr}(X) = \sum_{k=1}^n x_{kk} = \sum_{k=1}^n \underbrace{(x_{ek}, e_k)}_{\text{inner product}}$

Fact: $\text{Tr}(XY) = \text{Tr}(YX) \Rightarrow \text{Tr}(gXg^{-1}) = \text{Tr}(X)$, $g \in GL_n(\mathbb{K})$

So if $T = \mathcal{L}(V)$ (linear op. f.d. vs. V)

$$\text{Tr}(T) = \text{Tr}([T]_{\beta}), \beta \text{ any basis}$$

Def'n: let $\mathfrak{g}$ be a linear Lie algebra. $\rho: \mathfrak{g} \rightarrow \mathfrak{gl}(V)$

be a f.d.l. Lie algebra rep'n. Define:

$$B_{\rho}: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{K}, B_{\rho}(X, Y) = \text{Tr}(\rho(X), \rho(Y))$$

observe for x, y, z in $\mathfrak{g}$,

$$\begin{aligned} B_p([x, y], z) &= \text{Tr}((\rho(x)\rho(y) - \rho(y)\rho(x)) \cdot \rho(z)) \\ &= \text{Tr}(\rho(x)\rho(y)\rho(z) - \rho(y)\rho(x)\rho(z)) \\ &= \text{Tr}(\rho(x)\rho(y)\rho(z) - \rho(x)\rho(z)\rho(y)) \\ &= \text{Tr}(\rho(x) \cdot (\rho(y)\rho(z) - \rho(z)\rho(y))) \\ &= B_p(x, [y, z]) \end{aligned}$$

So we say that B_p is invariant in $\mathfrak{g}$.

In particular, let the Killing form

$$\begin{aligned} B(x, y) &= B_g(x, y) \\ &:= B_{\text{ad}}(x, y) = B(\text{ad}(x), \text{ad}(y)) \end{aligned}$$

Prop'n: let $\mathfrak{g}$ be a linear Lie algebra, $i \in \mathfrak{g}$ (Lie ideal).

then for $X \in \mathfrak{g}$, $Y \in \mathfrak{g}$, we have

$$\begin{aligned} (\text{Trace on } \mathfrak{L}(\mathfrak{g})) \quad \text{Tr}(\text{ad}_{\mathfrak{g}}(X), \text{ad}_{\mathfrak{g}}(Y)|_{\mathfrak{g}}) \\ = \text{Tr}(\text{ad}(X)\text{ad}(Y)) \quad (\text{Trace on } \mathfrak{L}(\mathfrak{g})). \end{aligned}$$

and hence $B_{\mathfrak{g}}|_{\mathfrak{g} \times \mathfrak{g}} = B_{\mathfrak{g}}$

Pf: let $\beta = (X_1, \dots, X_n)$ be a basis for $\mathfrak{g}$ s.t.

$\beta_{\mathfrak{g}} = (X_1, \dots, X_m)$ is a basis for $\mathfrak{g}$. Then

$$[\text{ad}(X)]_{\beta} = \begin{bmatrix} [\text{ad}_{\mathfrak{g}}(X)]_{\beta_{\mathfrak{g}}} & * \\ 0 & 0 \end{bmatrix}, \quad [\text{ad}(Y)]_{\beta} = \begin{bmatrix} [\text{ad}_{\mathfrak{g}}(Y)]_{\beta_{\mathfrak{g}}} & * \\ 0 & * \end{bmatrix}$$

$$\text{So } [\text{ad}(X)\text{ad}(Y)]_{\beta} = \begin{bmatrix} [\text{ad}_{\mathfrak{g}}(X)\text{ad}_{\mathfrak{g}}(Y)]_{\beta_{\mathfrak{g}}} & * \\ 0 & 0 \end{bmatrix}$$

Therefore the result follows.

□

Corollary (to Cartan's Criterion): if $\mathfrak{h} \leq \mathfrak{g}$ is a Lie subalgebra of some Lie algebra s.t. $B_{\mathfrak{g}}(x, y) = 0$ for $x, y \in \mathfrak{h}$, then $\mathfrak{h}$ is solvable.

Pf: By Cartan's Criterion, $\text{ad}_{\mathfrak{g}}(\mathfrak{h}) \subseteq \mathfrak{g}(\mathfrak{g})$ is solvable.

Since $\mathfrak{z}(\mathfrak{g}) = \ker(\text{ad}_{\mathfrak{g}})$ s.t.

$$\text{ad}_{\mathfrak{g}}(\mathfrak{h}) \cong \mathfrak{h} / \underbrace{(\mathfrak{h} \cap \mathfrak{z}(\mathfrak{g}))}_{\text{abelian ideal in } \mathfrak{h}}$$

which shows (earlier prop'n) that $\mathfrak{h}$ is solvable. $\square$.

Prop'n: let $\mathfrak{g}$ be a Lie algebra, $i \leq \mathfrak{g}$ (Lie ideal), then

$$i^B = \{x \in \mathfrak{g} : B(x, y) = 0 \text{ for all } y \in i\} \text{ ("B-annihilator")}$$

is an ideal in $\mathfrak{g}$.

Pf: Clearly, i^B is a linear subspace.

Let $X \in i^B$, $Y \in i$, $Z \in \mathfrak{g}$ and we have

$$B([X, Z], Y) = B(X, \underbrace{[Z, Y]}_{\in i}) = 0$$

Hence $i^B \triangleleft \mathfrak{g}$.

□.

Ex: On $\mathfrak{gl}_n(\mathbb{K})$, we have

$$B(X, Y) = 2n \operatorname{Tr}(XY) \rightarrow 2 \operatorname{Tr}(X) \operatorname{Tr}(Y). \quad (\text{trace on } L(\mathbb{K}^n) = \mathfrak{gl}_n(\mathbb{K}))$$

Hence on $\mathfrak{sl}_n(\mathbb{K})$ we have $B(X, Y) = 2n \operatorname{Tr}(XY)$

Notice that $[\mathfrak{gl}_n(\mathbb{K}), \mathfrak{gl}_n(\mathbb{K})] \subseteq \mathfrak{sl}_n(\mathbb{K}) \Rightarrow \mathfrak{sl}_n(\mathbb{K}) \triangleleft \mathfrak{gl}_n(\mathbb{K})$

Let $X, Y \in \mathfrak{gl}_n(\mathbb{K})$, compute

$$\text{ad}(X)\text{ad}(Y) E_{ij} = X(Y E_{ij} - E_{ij} Y) - (Y E_{ij} - E_{ij} Y) X$$

so with $(X, Y) = \text{Tr}(Y^* X)$, we have

$$\begin{aligned} (\text{ad}(X)\text{ad}(Y) E_{ij}, E_{ij}) &= \text{Tr}(E_{ji} X Y E_{ij} - E_{ji} X E_{ij} Y - E_{ji} Y E_{ij} X + E_{ji} E_{ij} Y X) \\ &= (XY)_{ii} - X_{ii} Y_{jj} - X_{jj} Y_{ii} + (YX)_{jj} \end{aligned}$$

$$\begin{aligned} \text{Then } B(X, Y) &= \text{Tr}(\text{ad}(X), \text{ad}(Y)) = \sum_{i,j=1}^n (\text{ad}(X)\text{ad}(Y) E_{ij}, E_{ij}) \\ &= \sum_{i,j=1}^n ((XY)_{ii} + (YX)_{jj} - X_{jj} Y_{ii} - X_{ii} Y_{jj}) \\ &= 2n \text{Tr}(XY) - 2\text{Tr}(X)\text{Tr}(Y) \end{aligned}$$

Ideals in $\mathfrak{gl}_n(\mathbb{K})$:

(1). Two ideals: $\mathcal{Z}(\mathfrak{gl}_n(\mathbb{K})) = \mathbb{K} I$, $\mathfrak{sl}_n(\mathbb{K})$

Notice that

$$\left\{ \underbrace{E_{ii} - E_{nn}}_{n-1} \ (i=1, \dots, n-1), \underbrace{E_{ij}}_{n^2-n} \ (i \neq j \in \{1, \dots, n\}) \right\}$$

is a basis for $\mathfrak{sl}_n(\mathbb{K})$. Hence $\dim(\mathfrak{sl}_n(\mathbb{K})) = n^2 - 1$

Thus $\mathfrak{gl}_n(\mathbb{K}) = \mathfrak{sl}_n(\mathbb{K}) \oplus \mathbb{K}I$.

(II) let $X \in \mathfrak{gl}_n(\mathbb{K})$ and $\langle X \rangle = \{ i \in \mathfrak{gl}_n(\mathbb{K}) : X \in i \}$

Recall

$$[E_{ij}, E_{kl}] = \begin{cases} E_{ii} - E_{jj} & , i=k, j=l \\ E_{il} & , j=k, i \neq l \\ -E_{kj} & , j \neq k, i=l \\ 0 & . \text{ o.w.} \end{cases}$$

This shows for $i \neq j$ that $\langle E_{ij} \rangle = \mathfrak{sl}_n(\mathbb{K})$.

Case #1: Say $X_{ij} \neq 0$ for some $i \neq j$.

Recall: $\langle X \rangle \triangleleft \mathfrak{gl}_n(\mathbb{K}) \Rightarrow \langle X \rangle$ is $\text{Ad}(GL_n(\mathbb{K}))$ -invariant.

$$\begin{aligned} \text{Ad}(\exp(E_{ji}))X &= \exp(E_{ji})X\exp(-E_{ji}) \\ &= (I + E_{ji})X(I - E_{ji}) \\ &= \underbrace{X}_{\in \langle X \rangle} + [E_{ji}, X] - E_{ji}X E_{ji} \end{aligned}$$

$$\Rightarrow \lambda_{ij} \bar{e}_{ij} \in \langle X \rangle \Rightarrow \mathfrak{sl}_n(K) = \langle \bar{e}_{ij} \rangle \subseteq \langle X \rangle$$

$$\Rightarrow \langle X \rangle = \mathfrak{sl}_n(K) \text{ or } \mathfrak{gl}_n(K).$$

Case #2: $X = \sum_{k=1}^n \lambda_{kk} \bar{e}_{kk}$. Then if $i \neq j$,

$$[\bar{e}_{ij}, X] = \sum_{k=1}^n \lambda_{kk} [\bar{e}_{ij}, \bar{e}_{kk}] = (\lambda_{jj} - \lambda_{ii}) \bar{e}_{ij}$$

Hence, as above, $\langle \bar{e}_{ij} \rangle \subseteq \langle X \rangle$ if $\lambda_{ii} \neq \lambda_{jj}$ for some $i \neq j$,

and $\langle X \rangle = K I$ otherwise.

$$\text{So } \langle X \rangle \in \{\mathfrak{sl}_n(K), \mathfrak{gl}_n(K), KI\}$$

(III) Conclusion: $\mathfrak{sl}_n(K), KI$ are the only proper ideals of $\mathfrak{gl}_n(K)$. Any ideal in $\mathfrak{sl}_n(K)$ must be an ideal in $\mathfrak{gl}_n(K)$,

and it follows that $\mathfrak{sl}_n(K)$ is simple.

$$\text{Also, } D^k(\mathfrak{gl}_n(K)) = \mathfrak{sl}_n(K), k \geq 2$$

$$D^k(\phi_n(k)) = \phi_n(k), \quad k \geq 1$$

Hence neither is solvable.

(IV) Real ideals in $\mathfrak{gl}(n, \mathbb{C})$.

Let $X \in \mathfrak{gl}(n, \mathbb{C}) \setminus \{0\}$, then, as above.

$$X_{ij} \bar{E}_{ji} \in \langle X \rangle_{\mathbb{R}} \quad (\text{smallest } \mathbb{R}\text{-ideal containing } X).$$

$$\Rightarrow \underset{\substack{\uparrow \\ |X_{ij}|}}{e^{it}} \bar{E}_{ji} \in \langle X \rangle_{\mathbb{R}} \text{ for some } t \in \mathbb{R}.$$

$$\text{if } s \in \mathbb{R}, \quad e^{is} \bar{E}_{ji} = \exp((s-t)E_{jj}) e^{it} \bar{E}_{ji} \exp((t-s)E_{jj})$$

$$= \text{Ad}(\exp(j(s-t)E_{jj})) \bar{E}_{ji}$$

$$\Rightarrow \alpha \bar{E}_{ji} \in \langle X \rangle_{\mathbb{R}} \text{ for any } \alpha \in \mathbb{C}.$$

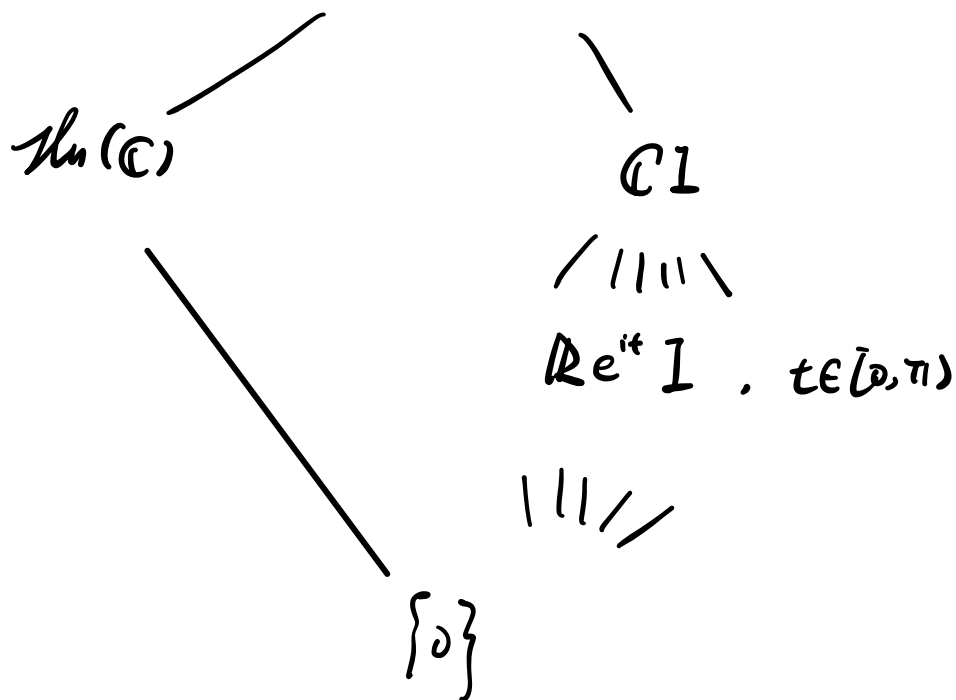
Similar in diagonal case.

Hence $\mathbb{R}$ -ideals are $\mathfrak{gl}(n, \mathbb{C}), \mathbb{R}e^{it}I, (t \in [0, \pi))$.

$$\mathbb{C}I = \mathfrak{z}(\mathfrak{gl}(n, \mathbb{C})). \quad \square$$

lie ideal lattice

$$\mathfrak{gl}_n(\mathbb{C}) = \mathfrak{gl}_n(\mathbb{R}) \oplus \mathbb{C}I$$



Prop'n: let $\mathfrak{g} \subseteq \mathfrak{gl}_n(\mathbb{C})$ be a $\mathbb{R}$ -lie subalgebra.

Then if $\mathfrak{g} + i\mathfrak{g} = \mathbb{C}\mathfrak{g}$ is simple.

then $\mathfrak{g}$ is simple.

Pf: if $\mathfrak{k} \triangleleft \mathfrak{g}$ ($\mathbb{R}$ -lie ideal), then $\mathfrak{k} + i\mathfrak{k} \triangleleft \mathfrak{g} + i\mathfrak{g}$ ($\mathbb{C}$ -lie ideal)

2.

Notation: With $\mathfrak{g}$ above, if $\mathfrak{g} \cap i\mathfrak{g} = \{0\}$, then we write

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} + i\mathfrak{g}$$

$$= \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$$

Example: $\mathfrak{u}(n) = \{X \in \mathfrak{gl}_n(\mathbb{C}) : \text{Tr}(X) = 0, X^* = -X\}$

is simple.

Basis: $X_{ij} = E_{ij} - E_{ji}$, $Y_{ij} = i(E_{ij} + E_{ji})$, $i < j \in [n]$.

$$H_k = i(E_{kk} - E_{k+1, k+1}), \quad k = 1, \dots, n-1$$

$$\Rightarrow \dim_{\mathbb{R}} \mathfrak{u}(n) = \underbrace{n^2 - n}_{X_{ij}, Y_{ij}} + n - 1 = n^2 - 1 = \dim_{\mathbb{C}} \mathfrak{u}(n, \mathbb{C})$$

Then $E_{ij} = \frac{1}{2}(X_{ij} - iY_{ij})$, $i < j$. $E_{kk} - E_{k+1, k+1} = -iH_k$

So $\mathfrak{u}(n) + i\mathfrak{u}(n) = \mathfrak{h}_n(\mathbb{C})$. Use prop'n.

Semisimple Lie algebras:

Def'n: let $\mathfrak{g}$ be a linear Lie algebra. We say that

• $\mathfrak{g}$ is simple if it admits no proper and $\dim_{\mathbb{F}} \mathfrak{g} \geq 2$.

• $\mathfrak{g}$ is semisimple if it admits no nonzero abelian ideals.

Note: (i) simple $\Rightarrow$ semisimple.

(ii) As $\ker \text{ad} = \mathfrak{z}(\mathfrak{g})$ (Centre), so if $\mathfrak{g}$ is semisimple,

$\text{ad}: \mathfrak{g} \rightarrow \mathcal{L}(\mathfrak{g})$ is injective.

(iii) If $\mathfrak{g}_1, \mathfrak{g}_2$ are semisimple, then

$$\mathfrak{g}_1 \oplus \mathfrak{g}_2 = \left\{ \begin{bmatrix} X & 0 \\ 0 & Y \end{bmatrix} : X \in \mathfrak{g}_1, Y \in \mathfrak{g}_2 \right\}$$

is semisimple. Indeed, if $\mathfrak{a} \triangleleft \mathfrak{g}_1 \oplus \mathfrak{g}_2$ is an abelian ideal, then $\mathfrak{a} \cap (\mathfrak{g}_1 \oplus \{0\}) \triangleleft \mathfrak{g}_1 \cong \mathfrak{g}_1 \oplus \{0\}$ is an abelian ideal, hence $\{0\}$. Similar with $\mathfrak{g}_2 \Rightarrow \mathfrak{a} = \{0\}$.

Remark: if $i, j \triangleleft \mathfrak{g}$, then $[i, j] \triangleleft \mathfrak{g}$.

If $X \in \mathfrak{g}, Y \in i, Z \in j$, then

$$[X, [Y, Z]] = \text{ad}(X)[Y, Z] = [\text{ad}(X)Y, Z] + [Y, \text{ad}(X)Z] \in [i, j].$$

Theorem: let $\mathfrak{g}$ be a linear Lie algebra. Then TFAE:

- (i) $\mathfrak{g}$ is semisimple
- (ii) $\text{rad}(\mathfrak{g}) = \{0\}$

(iii), the Killing form is non-degenerate: $\mathfrak{g}^B = \{0\}$.

Pf: (i) $\Rightarrow$ (ii): if $\mathfrak{r} = \text{rad}(\mathfrak{g}) \neq \{0\}$. Let $m = \min\{k \in \mathbb{N} : D^k(\mathfrak{r}) = \{0\}\}$.

Then $\mathfrak{a} = D^{m-1}(\mathfrak{r})$ is an abelian ideal in $\mathfrak{g}$ (last remark).

(ii) $\Rightarrow$ (iii): last corollary $\Rightarrow \mathfrak{g}^B$ is solvable.

Hence $\mathfrak{g}^B \subseteq \text{rad}(\mathfrak{g}) \subseteq \{0\}$.

(iii) $\Rightarrow$ (i): Were $\mathfrak{g}$ to have a nonzero abelian ideal $\mathfrak{a}$,

then for $X \in \mathfrak{a}$, $Y \in \mathfrak{g}$,

$$B(X, Y) = \text{Tr}(\text{ad}(X) \text{ad}(Y)) = \text{Tr}(\underbrace{\text{ad}_{\mathfrak{a}}(X)}_{=0} \text{ad}(Y)|_{\mathfrak{a}}) = 0$$

Thus $\mathfrak{a} \subseteq \mathfrak{g}^B$, violating assumption.

□.

Theorem (Structure of semisimple Lie algebra):

Let $\mathfrak{g}$ be a semi-simple linear Lie algebra. Then there

is a finite set of proper ideals, $\mathfrak{s}_1, \dots, \mathfrak{s}_n$ s.t.

- $\mathfrak{g} = \mathfrak{s}_1 \oplus \dots \oplus \mathfrak{s}_n$ - i.e., $\mathfrak{g} = \sum_{j=1}^n \mathfrak{s}_j$ and $\mathfrak{s}_i \cap \mathfrak{s}_j = \{0\}, i \neq j$
- $[\mathfrak{s}_i, \mathfrak{s}_j] = \{0\}, i \neq j$
- each $\mathfrak{s}_i$ is a simple Lie algebra

Moreover, $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$ (we call $\mathfrak{g}$ "perfect").

Pf: (I). if $0 \neq i \notin \mathfrak{g}$, then $\mathfrak{g} = i \oplus i^B$

Indeed, $i \cap i^B$ is an ideal of $\mathfrak{g}$ with vanishing Killing form.

Hence earlier corollary (Cartan's Criterion) shows it is solvable.

But (ii) above tells us that $i \cap i^B = \{0\}$. Let $j = i + i^B$.

As $i \subseteq j$, we have $j^B \subseteq i^B$. As $i^B \subseteq j$, $j^B \subseteq (i^B)^B$.

Thus $j^B \subseteq i^B \cap (i^B)^B = \{0\}$. By some argument we just

used in i . Now let $(Y_1, \dots, Y_m)$ be a basis for j .

Define $T: j \rightarrow K^n$, $T(X) = \begin{bmatrix} B(Y_1, X) \\ \vdots \\ B(Y_m, X) \end{bmatrix}$. So $\ker T = j^B = \{0\}$.

Thus $\dim_K j \leq m = \dim_K j \leq \dim_K g$. Hence $j = g$.

(II). Let s_1 be a non-zero ideal of g , of minimal dim.

if $s_1 = g$, we are done.

Otherwise, let s_2 be a nonzero ideal of s_1^B of minimal dim.

If $s_1 + s_2 = g$, notice $s_1 \cap s_2 = \{0\}$ by (I), we are done.

Otherwise, let s_3 be a nonzero ideal of $(s_1 + s_2)^B$ of min. dim.

Ultimately, we obtain ideals $\mathfrak{s}_1, \dots, \mathfrak{s}_n$ of $\mathfrak{g}$ so $(\mathfrak{s}_1 + \dots + \mathfrak{s}_n)^2 = \{0\}$.

$\mathfrak{s}_j \cap \mathfrak{s}_k = \{0\}$ for $j \neq k$, and $\mathfrak{g} = \mathfrak{s}_1 + \dots + \mathfrak{s}_n$, so

$$\mathfrak{g} = \mathfrak{s}_1 \oplus \dots \oplus \mathfrak{s}_n$$

Then each $[\mathfrak{s}_j, \mathfrak{s}_k] \subseteq \mathfrak{s}_j \cap \mathfrak{s}_k = \{0\}$.

If $i \in \mathfrak{s}_j$, then $i \in \mathfrak{g}$ since $\mathfrak{g} = \mathfrak{s}_1 + \dots + \mathfrak{s}_n$,

each $[\mathfrak{s}_k, \mathfrak{s}_j] = \{0\}$ for $k \neq j$.

So each $i \in \{0\}, \mathfrak{s}_j\}$ by min dim assumption on $\mathfrak{s}_j$.

Thus, $\mathfrak{s}_j$ is either simple or 1-dim'l, hence abelian.

But this violates Assumption.

$$\text{Finally, } [\mathfrak{g}, \mathfrak{g}] = \sum_{i,k=1}^n [\mathfrak{s}_i, \mathfrak{s}_k] = \sum_{j=1}^n [\mathfrak{s}_j, \mathfrak{s}_j] = \sum_{j=1}^n \mathfrak{s}_j = \mathfrak{g}. \quad \square$$

1

of: (i) affine motion

(ii) $G = GL_n(\mathbb{R})$, $g = g_{\mathbb{R}}(\mathbb{R})$, $\sigma = \text{id} \Rightarrow \lambda_g = L_g$

$$\int_{GL_n(\mathbb{R})} f = \int_{GL_n(\mathbb{R})} f([g_{ij}]) \frac{\prod_{i,j=1}^n dg_{ij}}{|\det [g_{ij}]|^n}$$

(iii) $G = GL_n(\mathbb{C})$. Hence $g = g_{\mathbb{C}}(\mathbb{C})$, and $\sigma = \text{id}$ so $\lambda_g = L_g$

Consider $GL_n(\mathbb{C}) \subseteq GL_{2n}(\mathbb{R})$, $[x_{k,l} + iy_{k,l}] \mapsto \begin{bmatrix} x_{k,l} & y_{k,l} \\ -y_{k,l} & x_{k,l} \end{bmatrix}$

We wish to compute $|\det L_g|$ for $g \in GL_n(\mathbb{C})$.

$$L_g: g_{\mathbb{C}}(\mathbb{C}) \rightarrow g_{\mathbb{C}}(\mathbb{C})$$

Jordan form provides $h \in GL_n(\mathbb{C})$ so

$$hgh^{-1} = \begin{bmatrix} \alpha_1 & & \\ & \ddots & \\ & & \alpha_n \end{bmatrix}$$

and hence

$$\tilde{h} \tilde{g} \tilde{h}^{-1} = \begin{bmatrix} a_{11} & b_{11} & & & \\ -b_{11} & a_{11} & & & \\ & & a_{22} & b_{22} & \\ & & -b_{22} & a_{22} & \\ 0 & & & & \ddots \end{bmatrix}$$

$$\begin{aligned} \text{So } |\det \tilde{g}| &= \prod_{k=1}^n |\det \begin{bmatrix} a_{kk} & b_{kk} \\ -b_{kk} & a_{kk} \end{bmatrix}| = \prod_{k=1}^n (a_{kk}^2 + b_{kk}^2) \\ &= \prod_{k=1}^n |\alpha_k|^2 = |\det g|^2 \end{aligned}$$

As with $GL_n(\mathbb{R})$ above, we get $|\det g| = |\det \tilde{g}|^n = |\det g|^{2n}$

Hence for $f \in C_c(GL_n(\mathbb{C}))$,

$$\int_{GL_n(\mathbb{C})} f = \int_{GL_n(\mathbb{C})} f(\bar{1}x_{k,l} + iy_{k,l}) \frac{\prod_{k,l=1}^n dx_{k,l} dy_{k,l}}{|\det \bar{1}x_{k,l} + iy_{k,l}|^{2n}}.$$

$$(i') \quad F(\mathbb{C}) = \left\{ \begin{bmatrix} \alpha & \beta \\ 0 & 1 \end{bmatrix} : \alpha \in \mathbb{C} \setminus \{0\}, \beta \in \mathbb{C} \right\}$$

Similarly (i), we get

$$\int_{f(\mathbb{C})} f = \int_{\mathbb{C}} f \left(\begin{bmatrix} a_1 + ia_2 & b_1 + ib_2 \\ 0 & 1 \end{bmatrix} \right) \frac{da_1 da_2 db_1 db_2}{(a_1^2 + a_2^2)^2}$$

(iv). $G = T_n^\circ(\mathbb{R})$ (1's on diagonal), $\sigma: \mathfrak{t}_n^\circ(\mathbb{R}) \rightarrow T_n^\circ(\mathbb{R})$.

$$\sigma(X) = I + X.$$

If $g \in T_n^\circ(\mathbb{R})$, $g = I + Y$, $Y \in \mathfrak{t}_n^\circ(\mathbb{R})$.

$$g\sigma(X) = (I + Y)(I + X) = I + (I + Y)X + Y = I + gX + Y$$

$$\text{So } \lambda_g(X) = \underbrace{gX}_{\text{linear}} + \underbrace{g - I}_{\text{trans}}.$$

$\frac{n(n-1)}{2} \times \frac{n(n-1)}{2}$ matrix

Hence $\det \lambda_g = \det (g|_{\mathfrak{t}_n^\circ(\mathbb{R})}) = 1$, as $[g|_{\mathfrak{t}_n^\circ(\mathbb{R})}]_\beta = [{}^1j_i^*]_\beta$
 β any basis for $\mathfrak{t}_n^\circ(\mathbb{R})$.

Thus. $\int_{T_0(\mathbb{R})} f = \int_{\mathbb{R}^n(\mathbb{R})} f\left(\begin{bmatrix} 1 & x_{1j} \\ & \ddots & \ddots \\ 0 & \ddots & 1 \end{bmatrix}\right) \prod_{1 \leq i < j \leq n} dx_{ij}.$

Towards a general theory of left-invariant integration

Let $G \subseteq GL_n(\mathbb{R})$ be a matrix Lie group.

For g in G , define the tangent plane to G at g by

$$T_g(G) = \left\{ \gamma'(0) \mid \gamma: \underbrace{(-\delta, \delta)}_{\substack{\text{Some interval} \\ \text{around } 0}} \rightarrow G, \gamma \text{ is diff'ble}, \gamma(0) = g \right\}$$

Proposition: $T_g(G) = g\mathfrak{g} = \mathfrak{g}g$

Pf: If $X \in \mathfrak{g}$, $gX = \frac{d}{dt} \underbrace{\gamma \exp(tX)}_{\gamma(t)} \Big|_{t=0} \in T_g(G)$, so $g\mathfrak{g} \subseteq T_g(G)$

Conversely, if $\delta: (-\delta, \delta) \rightarrow G$ is differentiable (on whole interval) with $\delta(0) = g$, we may assume that $\delta > 0$ is small enough so

$\|g^{-1}\delta(t) - 1\| < 1$, for $t \in (-\delta, \delta)$. Then define

$$\chi: (-\delta, \delta) \rightarrow \mathfrak{g}, \chi(t) = \log(g^{-1}\delta(t))$$

then $\chi(0) = 0$, $\chi(t) \in \mathfrak{g}$, so $\chi'(0) \in \mathfrak{g}$. Then let

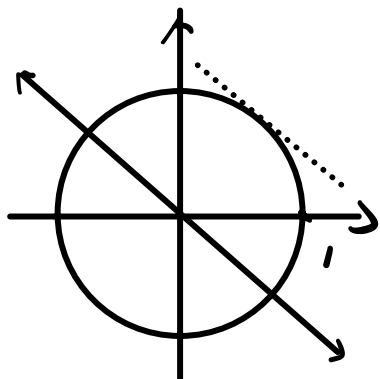
$$\delta(t) = g \exp(\log(g^{-1}\delta(t))) = g \exp(\chi(t)), \text{ and hence}$$

$$\delta'(0) = g \chi'(0) \in \mathfrak{g}g$$

Since g is Ad-invariant, $ggg^{-1} = g$, so $gg = gg$.

□.

eg: let $G = U(1) \subseteq M_1(\mathbb{C})$, i.e., $G = T \subseteq \mathbb{C}$.



$$\mathfrak{g} = \text{Lie}(U(1)) \cong i\mathbb{R}.$$

Fix notation (until Friday)

$G \subseteq GL_n(\mathbb{R})$ $\mathbb{R}$ -matrix Lie group, $\mathfrak{g} = \text{Lie}(G)$, $d = \dim_{\mathbb{R}} \mathfrak{g}$

Left invariant form on $\bigcup_{g \in G} g\mathfrak{g}$ "tangent bundle"

Fix $\omega_1 \in \wedge^d_{+}(\mathfrak{g})$ so $\omega_1 \neq 0$.

- if $x_2, \dots, x_d \in \mathfrak{g}$, $x \mapsto \omega_1(x, x_2, \dots, x_d)$ is linear on $\mathfrak{g}$
- if $x_1, \dots, x_d \in \mathfrak{g}$, $\sigma \in \text{Sym}(n)$, $\omega_1(x_{\sigma(1)}, \dots, x_{\sigma(d)}) = \text{sgn}(\sigma) \omega_1(x_1, \dots, x_d)$

g : let $\omega_1(x_1, \dots, x_d) = \det[\bar{[x_1]}, \dots, \bar{[x_d]}]$ (determinant of matrix of columns w.r.t. some basis)

Note: Up to scalar in $\mathbb{R} \setminus \{0\}$, this is unique.

Define for g in G , $\omega_g \in \Lambda^d(gg)$, by

$$\omega_g(gx_1, \dots, gx_d) = \omega_1(x_1, \dots, x_d) \quad (\text{L.I.})$$

Smooth vector fields: let G have C^1 -chart (C^∞ or analytic)

$\{(\varphi_i, U_i)\}_{i \in I}$ (each $\varphi_i: U_i \rightarrow \varphi_i(U_i) \subseteq \mathbb{R}^d$, $G = \bigcup_{i \in I} U_i$).

$\varphi_i \circ \varphi_j^{-1}: \varphi_i(U_i) \cap \varphi_j(U_j) \rightarrow \mathbb{R}^d$ are C^1 (C^∞ -analytic)

which is equivalent to the usual logarithmic chart (coordinate system)

Feb 26th

$$G \subseteq GL_n(\mathbb{R}), \quad g = \text{Lx}(G) = T_1(G), \quad d = \dim_{\mathbb{R}} g.$$

$$\omega_1 : g^d \rightarrow \mathbb{R}, \quad d\text{-linear alternating form ("determinant")}$$

$$\omega_g : (gg)^d \rightarrow \mathbb{R}, \quad \omega_g(gx_1, \dots, gx_d) = \omega_1(x_1, \dots, x_d) \quad (\text{L.I.})$$

$$C^\infty(G) : C^\infty\text{-functions..}$$

$$\Omega^\infty(G) = \{ \xi : G \rightarrow M_n(\mathbb{R}), \mid \xi_{i,j} \in C^\infty(G) \text{ each } i,j. \}$$

$$\{ \xi g, \in gg \text{ for } g \text{ in } G \}$$

" $T_1(G)$

— C^∞ -vector fields on G

Remark: $\Omega^\infty(G)$ is a $C^\infty(G)$ -module.

left-invariant fields: $X \in \mathfrak{g}$, let $P_X(g) = gX$, so $P_X \in \Omega^\infty(G)$

Now fix a basis $X_1, \dots, X_d$ for $\mathfrak{g}$, and let $(\cdot, \cdot)$

be the inner product which follows $X_1, \dots, X_d$ to be

an o.n.b.. If $\xi \in \Omega^\infty(G)$, write $\xi(g) \in \text{span}_{\mathbb{R}} \{gX_1, \dots, gX_d\}$

So $\xi(g) = \sum_{j=1}^d \underbrace{\xi_j(g)}_{\mathbb{R}\text{-coeff}} gX_j$. Then each $\xi_j(g) = (g^{-1}\xi(g), X_j)$

which shows that $\xi_j \in C^\infty(G)$. Thus

$$\xi = \sum_{j=1}^d \xi_j(\cdot) P_{X_j} \quad - \{P_{X_1}, \dots, P_{X_d}\} \text{ generate } \Omega^\infty(G) \\ \text{as a } C^\infty(G)\text{-module.}$$

Alternating d-forms $\Omega^\infty(G)$: Define

$$\omega: \Omega^\infty(G)^d \rightarrow C^\infty(G)$$

$$\omega(\xi_1, \dots, \xi_d)(g) = \omega_g(\xi_1(g), \dots, \xi_d(g))$$

If each $\xi_i = \sum_{j=1}^d \xi_{ij}(\cdot) p_{x_j}$ (as above), then for $g \in G$.

$$\omega(\xi_1, \dots, \xi_d)(g) = \omega_g\left(\sum_{j=1}^d \xi_{1j}(g) g x_j, \dots, \sum_{j=1}^d \xi_{dj}(g) g x_j\right)$$

$$= \det[\xi_{ij}(g)] \omega_g(g x_1, \dots, g x_d)$$

$$= \det[\xi_{ij}(g)] \omega_1(x_1, \dots, x_d).$$

If also $h \in G$,

$$\omega(\xi_1, \dots, \xi_d)(gh) = \det[\xi_{ij}(gh)] \omega_{gh}(gh x_1, \dots, gh x_d)$$

$$= \det[\xi_{ij}(gh)] \omega_h(h x_1, \dots, h x_d)$$

$$= \omega_h\left(\dots, \sum_{j=1}^d g^{-1} \xi_{ij}(gh) g x_j, \dots\right)$$

$$= \omega(g^{-1} \xi_1 \circ g, \dots, g^{-1} \xi_d \circ g)(h) \quad (\text{L.I.})$$

Notation: let $C_c(G) = \{f: G \rightarrow \mathbb{R} \mid f \text{ is cts. } \text{supp}(f) = \overline{f^{-1}(\mathbb{R} \setminus \{0\})} \text{ is compact}\}$

Goal: Define a left-invariant integral so

$$\int_G f = \int_G f \circ \gamma, \quad \gamma(h) = gh$$

Step 1: Sp. $\text{supp}(f) \subseteq V$, where (φ, U) is a C^∞ -coordinate patch.

Then $\varphi^{-1}: \underbrace{\varphi(U)}_{\text{open}} \subset \mathbb{R}^d \longrightarrow U \subset \mathcal{M}_n(\mathbb{R})$ is diff'ble.

$$\frac{\partial \varphi^{-1}}{\partial x_j}(x) = \frac{d}{dt} \underbrace{\varphi^{-1}(x_1, \dots, x_j + t, \dots, x_d)}_{\substack{G\text{-valued smooth curve} \\ \text{at } t=0 \text{ is } \varphi^{-1}(x)}} \Big|_{t=0} \in T_{\varphi^{-1}(x)}(G) = \varphi^{-1}(x)g \quad \text{for each } j.$$

Hence $\frac{\partial \varphi^{-1}}{\partial x_j} \circ \varphi \in \Omega^\infty(U)$ (vector field restricted to $U \subset G$)

Thus we define

$$\int_G f |\omega| := \int_{\varphi(U)} f(\varphi^{-1}(x)) \left| \omega \left(\frac{\partial \varphi^{-1}}{\partial x^1} \circ \varphi, \dots, \frac{\partial \varphi^{-1}}{\partial x^d} \circ \varphi \right) (\varphi^{-1}(x)) \right| dx$$

(Riemann int.
in $\varphi(U) \subset \mathbb{R}^d$)

$$= \int_{\varphi(U)} f \circ \varphi^{-1}(x) \left| \omega_{\varphi^{-1}(x)} (D\varphi^{-1}(x)) \right| dx$$

(forgive this
sleight-of-hand)

let's see that this is independent of coordinate patch.

let us suppose $\text{supp}(f) \subseteq U \cap V$ where (ψ, V) is another

C^∞ -coordinate patch.

$$\int_G f |\omega| = \int_{\varphi(U)} f \circ \varphi^{-1}(x) \left| \omega_{\varphi^{-1}(x)} D\varphi^{-1}(x) \right| dx$$

$$= \int_{\varphi(U)} f \circ \psi^{-1} \circ \psi \circ \varphi^{-1}(x) \left| \omega_{\varphi^{-1}(x)} D\varphi^{-1}(x) \right| dx$$

$$\overbrace{\quad}^{:= y}$$

$$\text{change of variable} \int_{\psi(U)} f \circ \psi^{-1}(y) |\omega_{\psi^{-1}(x)}| |D\psi^{-1}(x)| \frac{dy}{|\det D(\psi \circ \psi^{-1})(x)|}$$

$$= \int_{\psi(U)} f \circ \psi^{-1}(y) |\omega_{\underbrace{\psi^{-1}(x)}_{= \psi^{-1}(y)}}| (D_{\underbrace{\psi^{-1}(x)}_{= \psi^{-1}(y)}} \circ D(\psi \circ \psi^{-1})(y))| dy$$

$\uparrow$
 $x = \psi(y) = \psi(\psi^{-1}(y))$

$$= \int_{\psi(U)} f \circ \psi^{-1}(y) |\omega_{\psi^{-1}(y)}| |D\psi^{-1}(y)| dy \quad (\text{chain rule})$$

Thus, this is independent of equivalent coordinates.

lemma: (i) let $K \subseteq U \subseteq G$, K compact, U open. Then

there is f in $C_c(G)$ satisfying

$$f(G) \subseteq [0,1], \quad f|_K = 1, \quad \text{supp } f \subseteq U.$$

(ii) (Partition of unity) let $K \subseteq G$ be compact, $\{U_1, \dots, U_n\}$

be an open cover of K . Then there are $\{f_1, \dots, f_n\} \in C_c(G)$

- each $f_j \geq 0$

- $\text{supp}(f_j) \subseteq U_j$

- $(f_1 + \dots + f_n)|_K = 1$

} "partition of unity for K
subordinate to $\{U_1, \dots, U_n\}$ "

Pf: (i) $G \subseteq GL_n(\mathbb{R})$ is a metric space (use norm)

Thus for closed $F \subseteq G$,

$$j \mapsto \text{dist}(j, F) = \inf \{ \|j - h\| : h \in F \}$$

is cts (Lipschitz). let $r = \min \{ \text{dist}(j, K) : j \in G \setminus U \}$

and $V = \{ x \in G : \text{dist}(x, K) < \frac{r}{2} \}$ so U is open.

$$\text{let } f_j = \min \left\{ 1, \frac{\text{dist}(j, G \setminus U)}{r} \right\}.$$

$$(ii). \text{ let } k_j = K \setminus \bigcup_{\substack{i=1 \\ i \neq j}}^n U_i. \text{ If } k_j \neq \emptyset, \text{ find } \tilde{f}_j \in C_c(G)$$

$$\text{s.t. } \tilde{f}_j|_{k_j} = 1, \text{ and } \text{supp } \tilde{f}_j \subseteq U_j. \tilde{f}_j \geq 0.$$

$$\text{If } k_j = \emptyset, \text{ let } \tilde{f}_j = 0.$$

$$\text{let } U_j = \tilde{f}_j^{-1}(\mathbb{R} \setminus \{0\}), \text{ let } U = \bigcup_{j=1}^n U_j$$

$$\text{let } \tilde{f} \text{ s.t. } \tilde{f} \geq 0, \tilde{f}|_{\text{supp } \tilde{f}_1 \cup \dots \cup \text{supp } \tilde{f}_n} = 1, \text{supp } \tilde{f} \subseteq U.$$

$$\text{let } f_j = \frac{\tilde{f}_j}{\tilde{f}_1 + \dots + \tilde{f}_n} \tilde{f}.$$

□.

Friday, Feb 28

ERRATUM:

$$\begin{array}{ccc} & k \subseteq \bigcup_{j=1}^n V_j & \\ \uparrow & & \uparrow \\ \text{compact} & & \text{open} \end{array}$$

[if we let $k_j = k \setminus \bigcup_{\substack{i=1 \\ i \neq j}}^n V_i \leadsto$ we might get junk].

Instead, let each $V_{jr} = \{g \in V_j : \text{dist}(g, G \setminus V_j) > r\}$ so

$$k \subseteq \bigcup_{j=1}^n \bigcup_{r>0} V_{jr} \Rightarrow k \subseteq \bigcup_{j=1}^n V_{jr} \text{ for some } r>0$$

let $k_j = k \cap \overline{V_{jr}}$ for $j=1, \dots, n$. Proceed.

Left Invariant Integral: $f \in C_c(G)$

Step #1: $\text{supp } f \subseteq V$, (φ, U) coordinate patch.

$$\int_G f(\omega) = \int_{\varphi(U)} f(\varphi^{-1}(x)) \left| \omega_{\varphi^{-1}(x)} \left(\dots, \frac{\partial \varphi^i}{\partial x_j}(x), \dots \right) \right| dx$$

$\uparrow$
 open
 in $\mathbb{R}^d$

(Riemann integral in $\mathbb{R}^d$)

$\leadsto$ defined indep. of
equiv. coordinate.

Step #2: let $f \in C_c(G)$, $K = \text{supp}(f)$, $\{(\varphi_\alpha, U_\alpha)\}_{\alpha \in A}$

a C^∞ -chart (coordinate system) for G .

equivalent to usual one.

let $\{U_1, \dots, U_n\}$ be a finite cover of K ,

and let $\{f_1, \dots, f_n\}$ be a partition of unity for K .

subordinate to $\{U_1, \dots, U_n\}$.

Define $\int_G f(\omega) = \sum_{j=1}^n \int_G f f_j(\omega) = \sum_{j=1}^n \int_{\varphi_j(U_j)} (f f_j)(\varphi_j^{-1}(x)) \left| \omega_{\varphi_j^{-1}(x)} (D\varphi_j^{-1}(x)) \right| dx.$

let $\{h_1, \dots, h_n\}$ be any other p.o.v. subordinate to

$$\{h_1, \dots, h_n\}, \text{ then } f = \sum_{j=1}^n f f_j = \sum_{k=1}^n \sum_{j=1}^n f f_j h_k$$

$$= \sum_{j=1}^n \sum_{k=1}^n f h_k f_j = \sum_{k=1}^n f h_k.$$

So the integral is indep. of p.o.v.

Step #3: (left invariance)

let $f \in C_c(G)$, $g \in G$. We may assume that we are in step #1, so $\text{supp}(f \circ g) \subseteq U$, (φ, U) is a coordinate patch.

$$\int_G f \circ g |w| = \int_{\varphi(U)} f(g \varphi^{-1}(x)) \left| w_{\varphi^{-1}(x)} \left(\dots, \frac{\partial \varphi^{-1}}{\partial x_j}(x), \dots \right) \right| dx$$

$$= \int_{\varphi(U)} f(g \varphi^{-1}(x)) \underbrace{\left| w_x \left(\dots, \varphi^{-1}(x)^{-1} \frac{\partial \varphi^{-1}}{\partial x_j}(x), \dots \right) \right|}_{\varphi^{-1} g^{-1} g \frac{\partial \varphi^{-1}}{\partial x_j}(x)} dx \quad (\text{L.I.})$$

$$= \int_{\varphi(U)} f(\varphi^{-1}(x)) \left| \omega_{\varphi^{-1}(x)} \left(\dots, \frac{\partial \varphi^{-1}}{\partial x_j}(x), \dots \right) \right| dx \quad (\text{L.I.})$$

Let $\psi = \varphi \circ \psi^{-1}$, so $\psi^{-1} = \psi \circ \varphi^{-1}$, so my last expression is

$$= \int_{\psi(V)} f(\psi^{-1}(x)) \left| \omega_{\psi^{-1}(x)} \left(\dots, \frac{\partial \psi^{-1}}{\partial x_j}(x), \dots \right) \right| dx$$

$$= \int_G f |w|.$$

Prop'n (Right translation) For $f \in C_c(G)$, $g \in G$

$$\int_G f \circ R_g |w| = |\det \text{Ad}(g)| \int_G f |w|, \text{ where } R_g(h) = hg$$

Pf: Sp. $\text{supp}(f \circ R_g) \subseteq U$, (φ, U) a coordinate patch.

$$\int_G f \circ R_g |w| = \int_{\varphi(U)} f(\varphi^{-1}(x)) \left| \omega_{\varphi^{-1}(x)} \left(\dots, \frac{\partial \varphi^{-1}}{\partial x_j}(x), \dots \right) \right| dx$$

$$\begin{aligned}
&= \int_{\varphi(U)} f(\varphi^{-1}(x)g) \left| \omega_1 \left(\dots, \varphi^{-1}(x) \frac{\partial \varphi^{-1}}{\partial x_j}(x), \dots \right) \right| dx \\
&= \int_{\varphi(U)} f(\varphi^{-1}(x)g) \left| \omega_1 \left(\dots, \underbrace{g \varphi^{-1}(x) \frac{\partial}{\partial x_j} [\varphi^{-1}(x)g]}_{\in \text{Ad}(g^{-1})g = g}, \dots \right) \right| dx \\
&= \int_{\varphi(U)} f(\varphi^{-1}(x)g) |\det \text{Ad}(g)| \left| \omega_1 \left(\dots, g^{-1} \varphi^{-1}(x) \frac{\partial}{\partial x_j} [\varphi^{-1}(x)g], \dots \right) \right| dx.
\end{aligned}$$

Let $\psi = \varphi \circ R_{g^{-1}}$ so $\psi^{-1} = R_g \circ \varphi^{-1}$ and the last expression becomes

$$\begin{aligned}
&= |\det \text{Ad}(g)| \int_{\psi(U_g)} f(\psi^{-1}(x)) \left| \omega_{\psi(x)} \left(\dots, \frac{\partial \psi^{-1}}{\partial x_j}(x), \dots \right) \right| dx \\
&= |\det \text{Ad}(g)| \int_G f(w).
\end{aligned}$$

□

Def'n: Let $\Delta: G \rightarrow (0, +\infty)$, $\Delta(g) = \frac{1}{|\det \text{Ad}(g)|}$. We call

this the modular function. It satisfies

$$\Delta(g) \int f \circ R_{g^{-1}} |\omega| = \int f |\omega|.$$

Notice that Δ is a cts HM.

Prop'n: In the following cases, G is unimodular, i.e., $\Delta=1$.

- (i) G is abelian
- (ii) G is compact.
- (iii) $\mathfrak{g} \in \text{Lie}(G)$ is semisimple
- (iv) $\mathfrak{g}$ is reductive and G is connected
- (v) $\mathfrak{g}$ is nilpotent and G is connected.

Pf: (i) $L_g = R_g$

- (ii) $\Delta(G)$ is compact subgroup of $(0, \infty)$
- (iii) Recall if $\pi: G \rightarrow GL(V)$ (V f.d.) is a cts HM, then

$$d\pi(\text{Ad}(g)X) = \text{Ad}(\pi(g)) d\pi(X) \quad g \in G, X \in \mathfrak{g}.$$

Apply this to $\text{Ad}: G \rightarrow GL(\mathfrak{g})$ and we get (using $d\text{Ad} = \text{ad}$)

$$\text{ad}(\text{Ad}(g)X) = \text{Ad}(\text{Ad}(g)) \text{ad}(X) = \text{Ad}(g) \text{ad}(X) \text{Ad}(g)^{-1}.$$

Hence, for the Killing form, we have

$$B(\text{Ad}(g)X, \text{Ad}(g)Y) = \text{Tr} \left(\underbrace{\text{Ad}(g) \text{ad}(X) \text{Ad}(g)^{-1}}_{\text{ad}(\text{Ad}(g)X)} \underbrace{\text{Ad}(g) \text{ad}(Y) \text{Ad}(g)^{-1}}_{\text{ad}(\text{Ad}(g)Y)} \right)$$

$$= \text{Tr}(\text{ad}(X) \text{ad}(Y))$$

$$= B(X, Y), \text{ i.e., } B \text{ is Ad-invariant.}$$

Represent $B(X, Y) = [X]^T [B] [Y]$ (w.r.t. some basis on $\mathfrak{g}$).

Then from Ad-invariance of B , we get

$$[\text{Ad}(g)]^T [B] [\text{Ad}(g)] = [B]$$

So we take determinant to get $(\det \text{Ad}(g))^2 \det[B] = \det[B]$.

Since $\mathfrak{g}$ is semisimple, B is non-degenerate, so $\det[B] \neq 0$.

hence $(\det \text{Ad}(g))^2 = 1 \Rightarrow |\det \text{Ad}(g)| = 1$.

(iv) Write $\mathfrak{g} = \mathfrak{z} \oplus \mathfrak{s}$, $\mathfrak{z}$ centre of $\mathfrak{g}$, $\mathfrak{s}$ is semi-simple.

Each of $\mathfrak{z}$, $\mathfrak{s}$ are ad-invariant, hence Ad-invariant

as G is connected, with $\text{Ad}(g)|_{\mathfrak{z}} = I$ (identity)

for g in G . Thus, w.r.t. some basis, each

$$[\text{Ad}(g)] = \begin{bmatrix} I & 0 \\ 0 & [\text{Ad}(g)|_{\mathfrak{s}}] \end{bmatrix}$$

Thus, $|\det \text{Ad}(g)| = |\det g|_{\mathfrak{s}}| = 1$ from (iii) above.

(v). Since G is connected, any g in G is of the form $g = \exp(X_1) \cdots \exp(X_n)$, $X_1, \dots, X_n \in \mathfrak{g}$. Hence

$$\begin{aligned} \text{Ad}(g) &= \text{Ad}(\exp X_1) \cdots \text{Ad}(\exp X_n) \\ &= \exp(\text{ad}(X_1)) \cdots \exp(\text{ad}(X_n)) \quad (\text{dAd} = \text{ad}) \end{aligned}$$

Then $\det(\text{Ad}(g)) = \prod_{j=1}^n \det \exp(\text{ad}(X_j))$

used in $\rightarrow$ $\prod_{j=1}^n e^{\text{Tr}(\text{ad}(X_j))}$ $\leftarrow$ nilpotent, by Engel's Thm
determinant shift,

$$= \prod_{j=1}^n e^0 = 1$$

Example: Abelian Lie algebras are both reductive and nilpotent.

(A non-connected grp) $G = \left\{ \begin{bmatrix} e^n & b \\ 0 & 1 \end{bmatrix} : n \in \mathbb{Z}, b \in \mathbb{R} \right\}$

Notice that G is closed in $T = \left\{ \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} : a, b \in \mathbb{R}, a > 0 \right\}$.

and hence closed in $GL_2(\mathbb{R})$.

Plainly, $\mathfrak{g} = \text{Lie}(G) = \mathbb{R}Y$, $Y = E_{12} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

This is 1-dim'l $\Rightarrow$ it is abelian.

Coordinates: $V_n = \left\{ \begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix} : b \in \mathbb{R} \right\}$, $\varphi_n: V_n \rightarrow \mathbb{R}$,

$$\varphi_n \left(\begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix} \right) = b$$

$$\varphi_n^{-1}(b) = \begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix} \Rightarrow \frac{\partial \varphi_n^{-1}}{\partial b} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = Y$$

let $\omega_1: \mathfrak{g} \rightarrow \mathbb{R}$, $\omega_1(bY) = b$ (linear form hence "alternating").

$$\begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix} Y = z^n Y, \text{ so } \omega_{\begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix}}(z^n Y) = \omega_1(Y) = 1.$$

For $f \in C_c(G)$, we have

$$\int_G f(w) = \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix}\right) \left| \omega_{\begin{bmatrix} z^n & b \\ 0 & 1 \end{bmatrix}}\left(\frac{\partial \varphi_n^{-1}}{\partial b}(b)\right) \right| db$$

$$\overset{1}{\varphi_n(\psi_n)}$$

$$\overbrace{= \gamma}$$

(*) key step $= \sum_{n \in \mathbb{Z}} \int_{\mathbb{R}} f\left(\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}\right) \frac{1}{2^n} \underbrace{\left| \omega_{\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}}(2^n \gamma) \right|}_{= \omega_2(1) = 1} db$

$$= \sum_{n \in \mathbb{Z}} \frac{1}{2^n} \int_{\mathbb{R}} f\left(\begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix}\right) db$$

Notice that

$$\text{Ad} \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} \gamma = \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2^{-n} & -2^{-n}b \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2^n \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2^{-n} & -2^{-n}b \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 2^n \\ 0 & 0 \end{bmatrix} = 2^n \gamma$$

So $\det \text{Ad} \begin{bmatrix} 2^n & b \\ 0 & 1 \end{bmatrix} = 2^n$. This is not unimodular.